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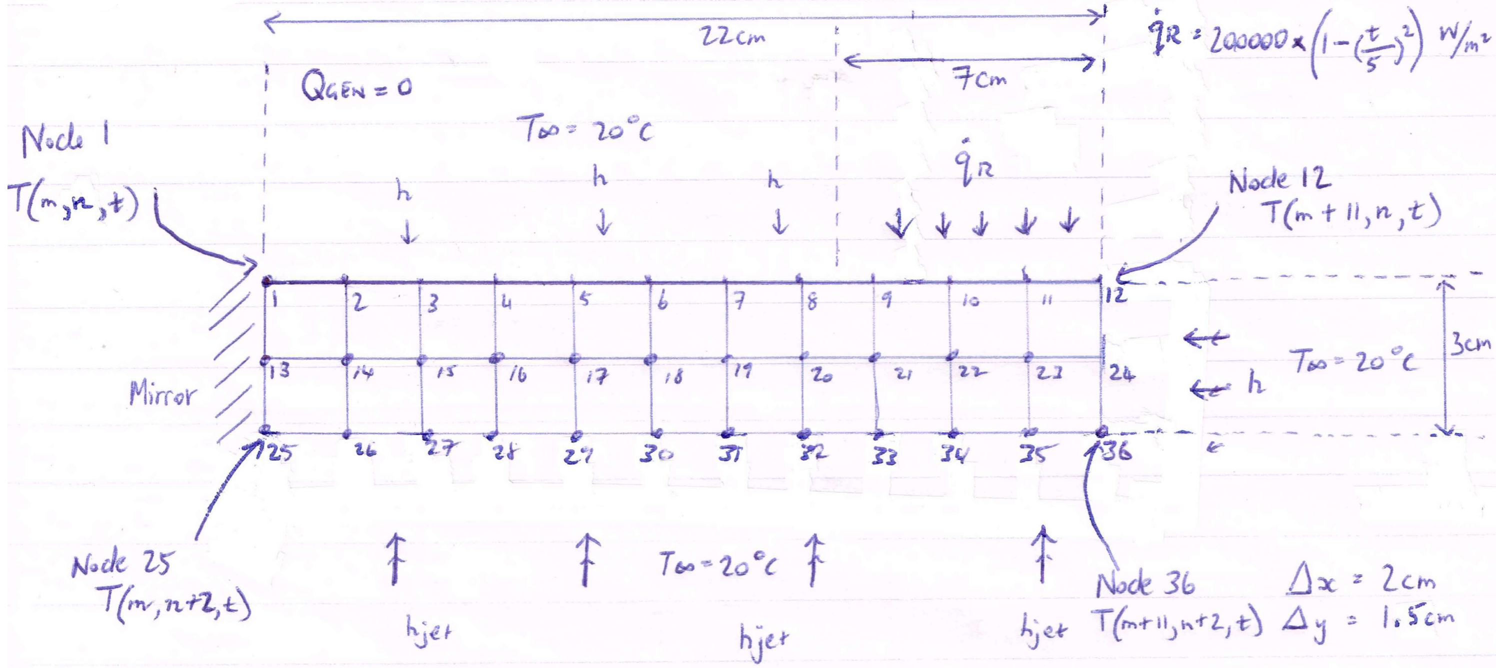
Numerical Transient 2D Heat transfer of Disk Brake calculations.



All nodes at 20°C initially. , Given: Material $\rho = 7000\ kg/m^3$, $c_p = 200\ J/kg\cdot K$, $k = 100\ W/m\cdot K$, $h_{jet} = 250\ W/m^2\cdot K$, $h = 25\ W/m^2\cdot K$

Finite Difference Nodal Equations ↓ / Assumptions: Transient 2D, Thermal conductivity constant, No internal heat gen, Radiation neglected.

Node ①

$$0 + h\frac{\Delta x}{2}(T_\infty - T_1) + k\frac{\Delta y}{2}\frac{T_2 - T_1}{\Delta x} + k\frac{\Delta x}{2}\frac{T_{13} - T_1}{\Delta y} = \rho\frac{\Delta x}{2}\frac{\Delta y}{2} c_p \frac{T_1^{l+1} - T_1}{\Delta t}$$

(q_{gen} removed, $=0$)

$$\times \frac{4}{k} \Rightarrow \frac{2h\Delta x}{k}(T_\infty - T_1) + 2\frac{\Delta y}{\Delta x}\left(T_2 - T_1\right) + 2\frac{\Delta x}{\Delta y}\left(T_{13} - T_1\right) = \frac{T_1^{l+1} - T_1}{\tau}$$

$$T_1^{l+1} = \left[\left(1 - \frac{2h\Delta x}{k}\tau - \frac{2\Delta y}{\Delta x}\tau - 2\frac{\Delta x}{\Delta y}\tau\right)T_1\right] + \tau\left[\frac{2h\Delta x}{k}T_\infty + 2\frac{\Delta y}{\Delta x}T_2 + 2\frac{\Delta x}{\Delta y}T_{13}\right]$$

Node ②

$$h\Delta x(T_\infty - T_2) + k\frac{\Delta y}{2}\frac{T_3 - T_2}{\Delta x} + k\Delta x\frac{T_{14} - T_2}{\Delta y} + k\frac{\Delta y}{2}\frac{T_1 - T_2}{\Delta x} = \rho\Delta x\frac{\Delta y}{2}c_p\frac{T_2^{l+1} - T_2}{\Delta t}$$

$$\times \frac{2}{k} \Rightarrow \frac{2h\Delta x}{k}(T_\infty - T_2) + \frac{\Delta y}{\Delta x}\left(T_3 - T_2\right) + 2\frac{\Delta x}{\Delta y}\left(T_{14} - T_2\right) + \frac{\Delta y}{\Delta x}\left(T_1 - T_2\right) = \frac{T_2^{l+1} - T_2}{\tau}$$

$$T_2^{l+1} = \left[\left(1 - \frac{2h\Delta x}{k}\tau - 2\frac{\Delta y}{\Delta x}\tau - \frac{2\Delta x}{\Delta y}\tau\right)T_2\right] + \tau\left[\frac{2h\Delta x}{k}T_\infty + \frac{\Delta y}{\Delta x}T_3 + 2\frac{\Delta x}{\Delta y}T_{14} + \frac{\Delta y}{\Delta x}T_1\right]$$

Node Node

③ → ⑧ as above form ↑

Node
(9)

$$\dot{q}_R \Delta x + k\frac{\Delta y}{2}\frac{T_{10}-T_9}{\Delta x} + k\Delta x \frac{T_{21}-T_9}{\Delta y} + k\frac{\Delta y}{2}\frac{T_8-T_9}{\Delta x} = \rho \Delta x \frac{\Delta y}{2} c_p \frac{T_9^{i+1}-T_9}{\Delta t}$$

$$\times \frac{2}{k} \Rightarrow \quad 2\dot{q}_R\frac{\Delta x}{k} + \frac{\Delta y}{\Delta x} T_{10}-T_9 + 2\frac{\Delta x}{\Delta y} T_{21}-T_9 + \frac{\Delta y}{\Delta x} T_8 - T_9 = \frac{T_9^{i+1}-T_9}{\tau}$$

$$T_9^{i+1} = \left[\left(1 - 2\frac{\Delta y}{\Delta x}\tau - 2\frac{\Delta x}{\Delta y}\tau\right)T_9\right] + \tau\left[2\dot{q}_R\frac{\Delta x}{k} + \frac{\Delta y}{\Delta x}T_{10} + 2\frac{\Delta x}{\Delta y}T_{21} + \frac{\Delta y}{\Delta x}T_8\right]$$

Node (10) + (11) as above form. ↑

Node (12) $\dot{q}_R \frac{\Delta x}{2} + h\frac{\Delta y}{2}(T_\infty - T_{12}) + k\frac{\Delta x}{2}\frac{T_{24}-T_{12}}{\Delta y} + k\frac{\Delta y}{2}\frac{T_{11}-T_{12}}{\Delta x} = \rho \frac{\Delta x}{2}\frac{\Delta y}{2} c_p \frac{T_{12}^{i+1}-T_{12}}{\Delta t}$

$$\times \frac{4}{k} \Rightarrow \quad 2\dot{q}_R\frac{\Delta x}{k} + 2h\frac{\Delta y}{k}(T_\infty - T_{12}) + 2\frac{\Delta x}{\Delta y}T_{24}-T_{12} + 2\frac{\Delta y}{\Delta x}T_{11}-T_{12} = \frac{T_{12}^{i+1}-T_{12}}{\tau}$$

$$T_{12}^{i+1} = \left[\left(1 - 2h\frac{\Delta y}{k}\tau - 2\frac{\Delta x}{\Delta y}\tau - 2\frac{\Delta y}{\Delta x}\tau\right)T_{12}\right] + \tau\left[2\dot{q}_R\frac{\Delta x}{k} + 2h\frac{\Delta y}{k}T_\infty + 2\frac{\Delta x}{\Delta y}T_{24} + 2\frac{\Delta y}{\Delta x}T_{11}\right]$$

Node (13)

$$k\Delta x\frac{T_1-T_{13}}{\Delta y} + k\Delta y\frac{T_{14}-T_{13}}{\Delta x} + k\Delta x\frac{T_{25}-T_{13}}{\Delta y} + k\Delta y\frac{T_{14}-T_{13}}{\Delta x} = \rho\Delta x\Delta y c_p\frac{T_{13}^{i+1}-T_{13}}{\Delta t}$$

$$\times \frac{1}{k} \Rightarrow \frac{\Delta x}{\Delta y}T_1 - T_{13} + \frac{\Delta y}{\Delta x}T_{14}-T_{13} + \frac{\Delta x}{\Delta y}T_{25}-T_{13} + \frac{\Delta y}{\Delta x}T_{14}-T_{13} = \frac{T_{13}^{i+1}-T_{13}}{\tau}$$

$$T_{13}^{i+1} = \left[\left(1 - 2\frac{\Delta x}{\Delta y}\tau - 2\frac{\Delta y}{\Delta x}\tau\right)T_{13}\right] + \tau\left[\frac{\Delta x}{\Delta y}T_1 + \frac{\Delta y}{\Delta x}T_{14} + \frac{\Delta x}{\Delta y}T_{25} + \frac{\Delta y}{\Delta x}T_{14}\right]$$

Node (14) $k\Delta x\frac{T_2-T_{14}}{\Delta y} + k\Delta y\frac{T_{15}-T_{14}}{\Delta x} + k\Delta x\frac{T_{26}-T_{14}}{\Delta y} + k\Delta y\frac{T_{13}-T_{14}}{\Delta x} = \rho\Delta x\Delta y\, c_p\frac{T_{14}^{i+1}-T_{14}}{\Delta t}$

$$\times 1/k \Rightarrow \frac{\Delta x}{\Delta y}T_2 - T_{14} + \frac{\Delta y}{\Delta x}T_{15}-T_{14} + \frac{\Delta x}{\Delta y}T_{26}-T_{14} + \frac{\Delta y}{\Delta x}T_{13}-T_{14} = \frac{T_{14}^{i+1}-T_{14}}{\tau}$$

$$T_{14}^{i+1} = \left[\left(1 - 2\frac{\Delta x}{\Delta y}\tau - 2\frac{\Delta y}{\Delta x}\tau\right)T_{14}\right] + \tau\left[\frac{\Delta x}{\Delta y}T_2 + \frac{\Delta y}{\Delta x}T_{15} + \frac{\Delta x}{\Delta y}T_{26} + \frac{\Delta y}{\Delta x}T_{13}\right]$$

Nodes (15) to (23) as above form ↑

Node 24

$$\frac{k\Delta x}{2}\,\frac{T_{12}-T_{24}}{\Delta y} + h\Delta y\,(T_\infty - T_{24}) + k\Delta x\,\frac{T_{36}-T_{24}}{\Delta y} + k\Delta y\,\frac{T_{23}-T_{24}}{\Delta x} = \rho\,\frac{\Delta x}{2}\,\Delta y\,c_p\,\frac{T_{24}^{i+1}-T_{24}}{\Delta t}$$

$$\times \frac{2}{k} \Rightarrow \frac{\Delta x}{\Delta y}(T_{12}-T_{24}) + \frac{2h\Delta y}{k}(T_\infty - T_{24}) + \frac{\Delta x}{\Delta y}(T_{36}-T_{24}) + \frac{\Delta y}{\Delta x}(T_{23}-T_{24}) = \frac{T_{24}^{i+1}-T_{24}}{\tau}$$

$$T_{24}^{i+1} = \left[\left(1 - \frac{\Delta x}{\Delta y}\tau - \frac{2h\Delta y}{k}\tau - \frac{\Delta x}{\Delta y}\tau - \frac{\Delta y}{\Delta x}\tau\right)T_{24}\right] + \tau\left[\frac{\Delta x}{\Delta y}T_{12} + \frac{2h\Delta y}{k}T_\infty + \frac{\Delta y}{\Delta x}T_{36} + \frac{\Delta y}{\Delta x}T_{23}\right]$$

Node 25

$$k\frac{\Delta x}{2}\,\frac{T_{13}-T_{25}}{\Delta y} + \frac{k\Delta y}{2}\,\frac{T_{26}-T_{25}}{\Delta x} + h_{jet}\frac{\Delta x}{2}(T_\infty - T_{25}) = \rho\,\frac{\Delta x}{2}\,\frac{\Delta y}{2}\,c_p\,\frac{T_{25}^{i+1}-T_{25}}{\Delta t}$$

$$\times \frac{4}{k} \Rightarrow 2\frac{\Delta x}{\Delta y}(T_{13}-T_{25}) + \frac{2\Delta y}{\Delta x}(T_{26}-T_{25}) + \frac{2h_{jet}\Delta x}{k}(T_\infty - T_{25}) = \frac{T_{25}^{i+1}-T_{25}}{\tau}$$

$$T_{25}^{i+1} = \left[\left(1 - \frac{2\Delta x}{\Delta y}\tau - \frac{2\Delta y}{\Delta x}\tau - \frac{2h_{jet}\Delta x}{k}\tau\right)T_{25}\right] + \tau\left[\frac{2\Delta x}{\Delta y}T_{13} + \frac{2\Delta y}{\Delta x}T_{26} + \frac{2h_{jet}\Delta x}{k}T_\infty\right]$$

Node 26

$$k\frac{\Delta x}{2}\,\frac{T_{14}-T_{26}}{\Delta y} + \frac{k\Delta y}{2}\,\frac{T_{27}-T_{26}}{\Delta x} + h_{jet}\Delta x(T_\infty - T_{26}) + \frac{k\Delta y}{2}\,\frac{T_{25}-T_{26}}{\Delta x} = \rho\,\Delta x\,\frac{\Delta y}{2}\,c_p\,\frac{T_{26}^{i+1}-T_{26}}{\Delta t}$$

$$\times \frac{2}{k} \Rightarrow \frac{\Delta x}{\Delta y}(T_{14}-T_{26}) + \frac{\Delta y}{\Delta x}(T_{27}-T_{26}) + \frac{2h_{jet}\Delta x}{k}(T_\infty - T_{26}) + \frac{\Delta y}{\Delta x}(T_{25}-T_{26}) = \frac{T_{26}^{i+1}-T_{26}}{\tau}$$

$$T_{26}^{i+1} = \left[\left(1 - \frac{\Delta x}{\Delta y}\tau - \frac{\Delta y}{\Delta x}\tau - \frac{2h_{jet}\Delta x}{k}\tau - \frac{\Delta y}{\Delta x}\tau\right)T_{26}\right] + \tau\left[\frac{\Delta x}{\Delta y}T_{14} + \frac{\Delta y}{\Delta x}T_{27} + \frac{2h_{jet}\Delta x}{k}T_\infty + \frac{\Delta y}{\Delta x}T_{25}\right]$$

T_{27} 15 28 ∞ 26

Node 27 → Node 35 as above form ↑

Node 36

$$k\frac{\Delta x}{2}\,\frac{T_{24}-T_{36}}{\Delta y} + \frac{h\Delta y}{2}(T_\infty - T_{36}) + h_{jet}\frac{\Delta x}{2}(T_\infty - T_{36}) + \frac{k\Delta y}{2}\,\frac{T_{35}-T_{36}}{\Delta x} = \rho\,\frac{\Delta x}{2}\,\frac{\Delta y}{2}\,c_p\,\frac{T_{36}^{i+1}-T_{36}}{\Delta t}$$

$$\times \frac{4}{k} \Rightarrow \frac{2\Delta x}{\Delta y}(T_{24}-T_{36}) + \frac{2h\Delta y}{k}(T_\infty - T_{36}) + \frac{2h_{jet}\Delta x}{k}(T_\infty - T_{36}) + \frac{2\Delta y}{\Delta x}(T_{35}-T_{36}) = \frac{T_{36}^{i+1}-T_{36}}{\tau}$$

$$T_{36}^{i+1} = \left[\left(1 - \frac{2\Delta x}{\Delta y}\tau - \frac{2h\Delta y}{k}\tau - \frac{2h_{jet}\Delta x}{k}\tau - \frac{2\Delta y}{\Delta x}\tau\right)T_{36}\right] + \tau\left[\frac{2\Delta x}{\Delta y}T_{24} + \frac{2h\Delta y}{k}T_\infty + \frac{2h_{jet}\Delta x}{k}T_\infty + \frac{2\Delta y}{\Delta x}T_{35}\right]$$